

IoT platform for real-time energy optimization in bioenergy plants, integrating pre-separator and conveyor data to reduce energy consumption and operational costs.

Energy × Energy optimisation × IoT platform · OS156 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
61 is the world moving	54 can we play, can we win	HIGH 0 named in the evidence	€8.0m partial evidence · per year	NOW budgeted procurement within 1...	L0 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is an IoT platform that optimizes energy consumption in bioenergy plants by integrating data from pre-separators and conveyor systems. It targets plant operators and energy managers who need to reduce operational costs and improve efficiency. The timing is driven by recent procurement activity in the sector and advances in edge AI and IoT frameworks.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€9.7m €1.5m – €150m	€8.0m €1.3m – €124m	€120k €19k – €1.9m	partial
Observed: tendered contract value, annualised	€172m €172m – €460m	€0	€0	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (NL+DK)	279 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting IoT platform	47.8 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	68 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Enterprises using IoT Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

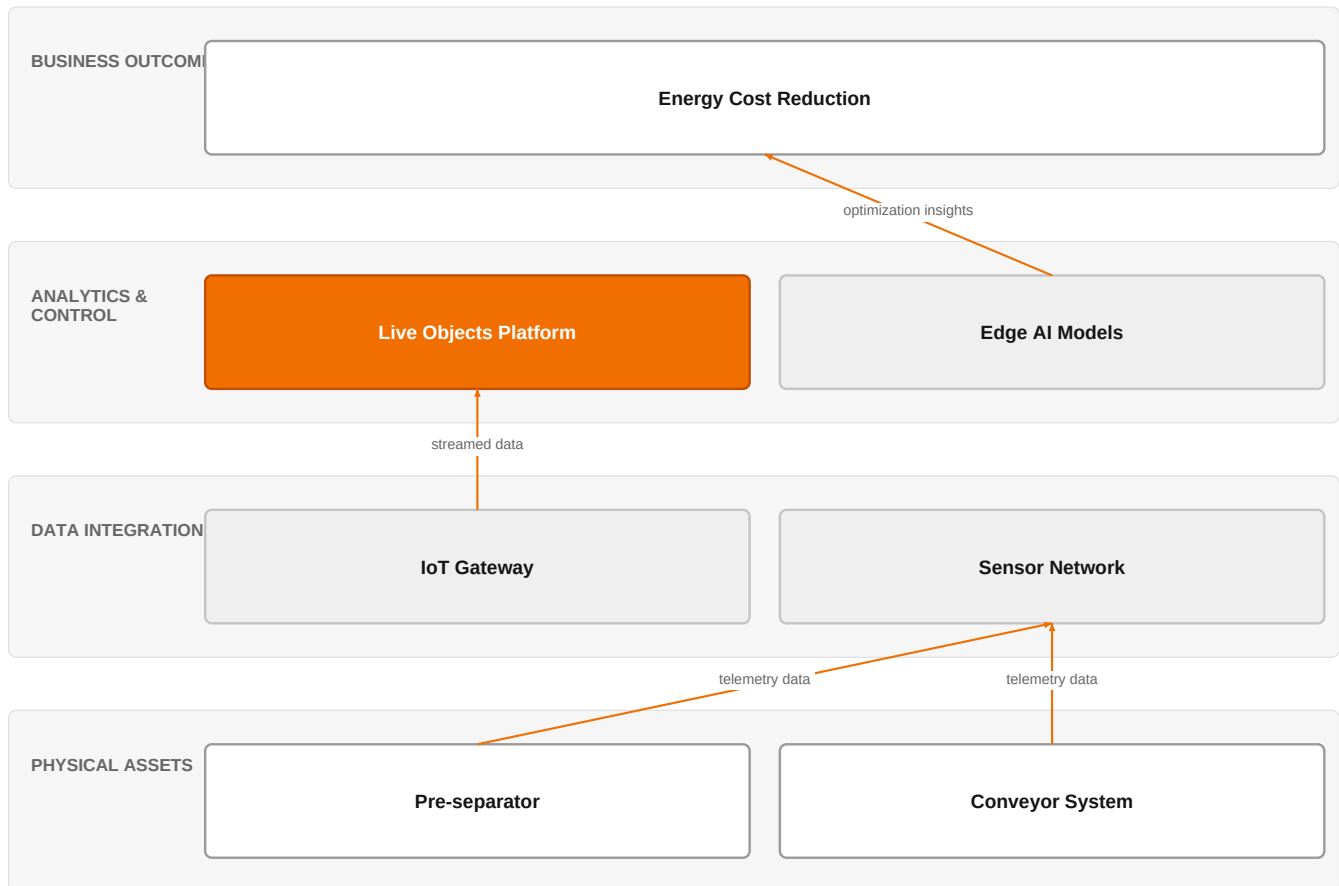
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

IoT Energy Optimization for Bioenergy Plants



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how sensor data from pre-separators and conveyors flows through an IoT platform with edge AI to deliver energy cost reductions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Bioenergy plants are actively procuring equipment such as pre-separators and conveyor belts, indicating investment in infrastructure that can be instrumented for energy optimization. Meanwhile, research demonstrates that edge-based IoT frameworks and AI can enable real-time monitoring and energy-efficient operations, reducing latency and power consumption. These developments make it feasible to deploy an integrated IoT platform that leverages sensor data for energy savings.

Sources: SIG-ED6B4C16F384 ted.europa.eu 2026-07-08; SIG-7A360EFB1A83 ted.europa.eu 2026-05-20; SIG-B5357D13A8A6 Sadhana 2026-06-15; SIG-F48074B55B20 openalex.org 2025-10-16

Who buys, and why now

The plant operations director or energy manager signs the contract, but the maintenance and process engineers feel the pain of high energy bills and inefficient equipment. The trigger is often an energy audit or a budget review that highlights rising costs, or a new regulatory requirement to reduce carbon footprint. They need to act because energy is a major operational expense and they lack visibility into real-time consumption across the plant.

Why it is hot now — every claim bound to a dated source

- Market consultations and procurement notices indicate bioenergy plants are investing in equipment like pre-separators and conveyor belts. [SIG-ED6B4C16F384, SIG-7A360EFB1A83]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deploy its Live Objects IoT platform to collect data from sensors on pre-separators and conveyors, using IoT experts to design and implement the solution. The engagement would start with a discovery workshop to map existing equipment and data sources, then integrate sensors and connect them to Live Objects. Orange Group researchers could contribute edge AI models for anomaly detection and energy optimization, and the solution would be hosted on SecNumCloud-certified infrastructure to meet sovereignty requirements.

Why Orange can win

Orange can win by combining its Live Objects platform with its IoT expertise and research capabilities, all delivered on sovereign, certified infrastructure (SecNumCloud). This combination is rare among competitors, who may offer platforms but lack the same level of trust and local support. However, the assets are not unique, and competitors like AWS or Deutsche Telekom have strong IoT offerings, so Orange must differentiate on security and vertical expertise.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells IoT platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	sells IoT platform; competes in OX: Smart Industries	structural	—
Proximus	Telecom operator peer	sells IoT platform	structural	—
Telefonica Tech	Telecom operator peer	sells IoT platform; active in Energy	structural	—
Telia Company	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
Verizon Business	Telecom operator peer	sells IoT platform	structural	—
Vodafone Business	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current energy consumption per ton of biomass processed, and how do you track it?
- 2 Do you have sensors on your pre-separator and conveyor systems, and if so, what data do they collect?
- 3 What is your biggest operational pain point: energy costs, downtime, or regulatory compliance?
- 4 Have you already evaluated any IoT or energy management platforms? If so, what was missing?
- 5 How would you describe your current data infrastructure – is it on-premise, cloud, or hybrid?
- 6 What is your timeline for making a decision on energy optimization investments?

Objections you will meet

They will say	You can answer
We already have a SCADA system that monitors our plant. Why do we need another platform?	That's a fair point. SCADA systems are great for control, but they often lack the advanced analytics and edge AI capabilities needed for predictive energy optimization. Our platform can complement your existing SCADA by adding a layer of intelligence that reduces energy consumption without replacing your current systems.
We are concerned about the security of IoT devices and data.	Security is a valid concern. Orange's solution is built on SecNumCloud-certified infrastructure and follows ISO 27001 standards, ensuring data sovereignty and protection. We also have deep experience in secure IoT deployments, so we can address your security requirements from day one.
How do we know the energy savings will justify the investment?	That's a critical question. We start with a pilot phase where we deploy sensors and analytics on a small scale to measure actual energy savings. This gives you concrete data to evaluate the ROI before a full rollout. Our experts will work with your team to define clear KPIs and benchmarks.

Risks and unknowns

The main risk is that bioenergy plants may already have legacy systems that are hard to integrate, or they may be cautious about new technology. The business case for energy savings may be uncertain without detailed plant data. Additionally, the competitive landscape is intense, and Orange's differentiation on sovereignty may not be a priority for all customers.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a bioenergy plant's energy optimization project, detailing how Live Objects can integrate pre-separator and conveyor data, and highlight Orange's IoT expertise and certifications (ISO 27001, SecNumCloud) to differentiate the proposal.
Sales	In a conversation with the operations director of a bioenergy plant, say: 'We can help you reduce energy consumption and operational costs by integrating your pre-separator and conveyor data in real time using our IoT platform, Live Objects, which is certified to ISO 27001 and SecNumCloud.'
Strategist / Innovator	Commission a deep-dive brief on how Orange's IoT experts and Group researchers can leverage Live Objects to integrate pre-separator and conveyor data for real-time energy optimization in bioenergy plants, focusing on the certification advantages of ISO 27001 and SecNumCloud.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-ED6B4C16F384	2026-07-08	ted.europa.eu	1	Netherlands – Heat-exchange units, air-conditioning and refrigerating equipment, and filtering ...
SIG-B5357D13A8A6	2026-06-15	Sadhana	1	CoMIEC: Configurable MQTT-based IoT edge computing framework for latency power trade-off
SIG-7A360EFB1A83	2026-05-20	ted.europa.eu	1	Denmark – Rubber conveyor belts – BIO_Transportband & ruller_26-31_FA
SIG-36DA075F2A6A	2026-03-02	openalex.org	1	IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism
SIG-216688A4B953	2026-03-01	openalex.org	1	Digital Twin-Enabled Thermal Energy Management System for Sustainable Manufacturing Process Opt...
SIG-03A062479220	2026-02-03	openalex.org	1	Robust Stochastic Power Allocation for Industrial IoT Federated Learning with Neurosymbolic AI
SIG-1BE0A6CA2278	2026-01-01	openalex.org	1	Reinforcement Learning-Guided Federated Multi-Task Learning for Adaptive Energy Management in I...
SIG-D8ADEDBAD839	2026-01-01	Elsevier BV	1	Robust Energy–Latency Co-Optimization for Power Allocation and Computation Offloading in Edge C...
SIG-395E0DE089C6	2025-11-12	openalex.org	1	Adaptive Federated Learning for Industrial Wireless Energy Optimization
SIG-C8282154F431	2025-10-28	openalex.org	1	Lightweight Signal Processing and Edge AI for Real-Time Anomaly Detection in IoT Sensor Networks
SIG-BCCCB520B0E1	2025-10-21	Institute of Electrical a...	1	Design and Integration of IoT-Enabled Health Sensors for Remote Patient Monitoring
SIG-D4C81A60CDDC	2025-10-17	openalex.org	1	An Energy-Aware Generative AI Edge Inference Framework for Low-Power IoT Devices
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-C1CD7564003C	2026-08-13	thisdaylive.com	2	AI , Digital Technology to Transform Energy Industry – THISDAYLIVE

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:32+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.